

Math for America minicourse

Introduction to Category Theory

Session 3: Universal properties

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Wednesday May 17, 2023

Plan

0. Recap of last session
1. Role vs character
2. Extremes
3. Uniqueness
4. Example: sets
5. Products
6. Context

0. Recap of last session

- We looked at more examples of categories: sets, ordered sets, groups
- We saw how category theory expresses a more nuanced notion of sameness called isomorphism, via invertible morphisms.
- We looked at isomorphisms of sets, ordered sets, and groups.

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Today:

We will look at how category theory characterizes things by the role they play in context.

1. Role vs character

In normal language we mix up role and intrinsic character

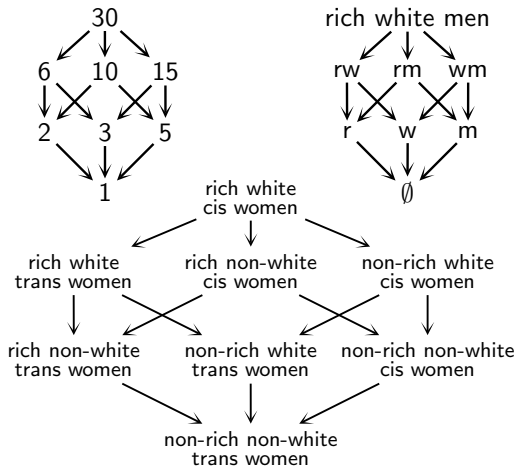
For example: film.

Think about role and intrinsic character for these examples.

- i. baseball cap
- ii. pumpkin spice
- iii. whipped cream
- iv. fat-free half-and-half
- v. teaspoon
- vi. pound cake
- vii. cookie, biscuit

2. Extremes

Last time we saw that these categories are
isomorphic.



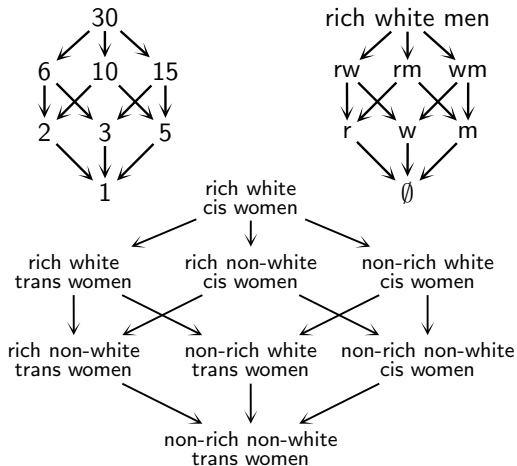
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play analogous roles within those contexts.

Can you describe their role, just by talking
about arrows in the category?

2. Extremes

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Category theory seeks to characterize things by the role they play in a category, just by describing interactions between arrows.

Advantage: we can go round looking for analogous situations in any category.

Another example: natural numbers

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longrightarrow \dots$$

2. Extremes: initial and terminal objects

We look for an object where all arrows “start”.

Definition

An **initial object** in a category $\mathcal{C}$ is an object I such that $\forall X \in \mathcal{C}, \exists ! I \rightarrow X$

translation:

For all X in $\mathcal{C}$, there exists a unique morphism $I \rightarrow X$

A **terminal object** in a category $\mathcal{C}$ is an object T such that $\forall X \in \mathcal{C}, \exists ! X \rightarrow T$

translation:

For all X in $\mathcal{C}$ there exists a unique morphism $X \rightarrow T$

Terminal objects are **dual** to initial objects:
the definitions are the same
just with all the arrows reversed.

Questions one might ask

- When are there terminal and initial objects?
- How many can there be? Are there ever none?
- Can something be both terminal and initial?

Thinking mathematically is as much about asking good questions as about finding answers.

Example

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longrightarrow \cdots$$

does not have a terminal object because:

It is “trying” to be: **infinity, but that's not a natural number**

2. Extremes: Ways to fail to have one

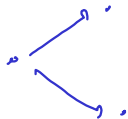
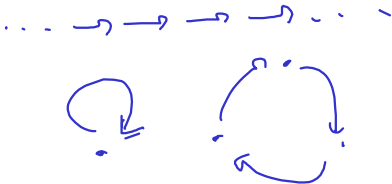
Explore: try and draw some categories which fail to have a terminal/initial object. Try and classify the types of failure into which features cause them to fail.

Something to try and prove: Can a category have more than one initial object? (Think about whether a room can have two “tallest people” and if so, what does that tell us about them?)

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infinite, loops, parallel, branching, disconnected, no arrows, no objects

Break to 6:39

3. Uniqueness

Theorem If I and I' are both initial then they are uniquely isomorphic.

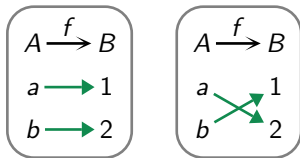
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We saw last time that things can be isomorphic in several different ways.

For example two 2-element sets are isomorphic in two ways:



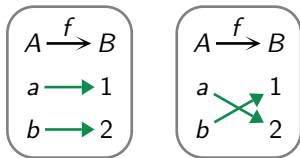
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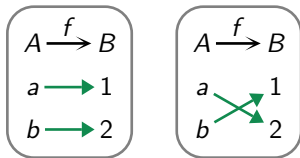
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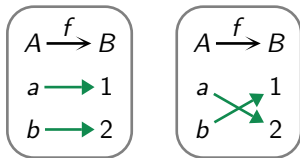
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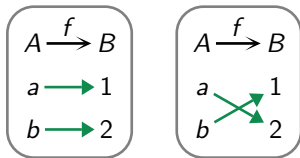
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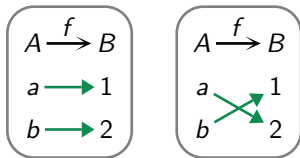
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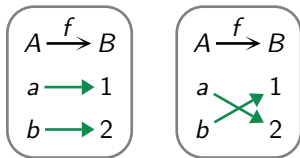
and it is the identity 1_I , so we know $g \circ f = 1_I$.

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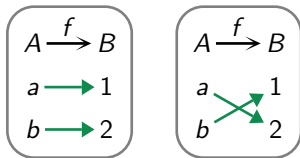
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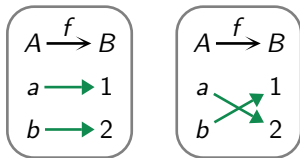
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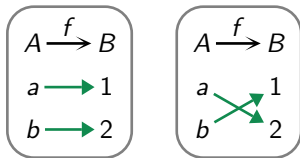
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Being an initial/terminal object is a type of **universal property**.

4. Example: sets

In the category of sets and functions.

How many functions $A \longrightarrow B$.

We write $|A|$ for the number of elements in A .

Number of functions $A \longrightarrow B$ is $|B|^{|A|}$.

What values of $|A|$ and $|B|$ would make this 1?

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What values of $|A|$ and $|B|$ would make this 1?

- If $|B| = 1$ then raising it to any power gives 1.

This tells us that if B has 1 object there is always exactly 1 function **to** it.

That is: any 1-element set is **terminal**.



- If $|A| = 0$ then the formula always gives 1.

This tells us that if A has 0 objects then there is always exactly 1 function **out of** it.

That is: the empty set $\emptyset$ is **initial**.

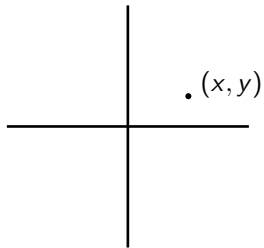
How is this possible??

The definition of being a function is **vacuously** satisfied.

5. Products

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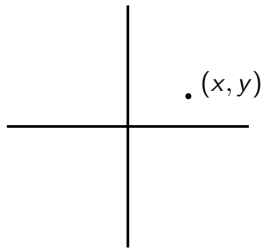
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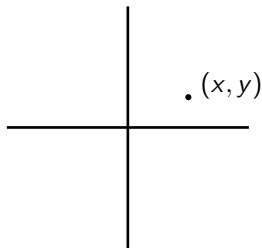
Example:

$$A = \{1, 2, 3\}, \quad B = \{a, b\}$$

b	$(1, b)$	$(2, b)$	$(3, b)$
a	$(1, a)$	$(2, a)$	$(3, a)$
	1	2	3

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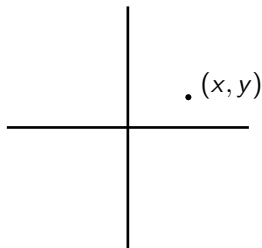
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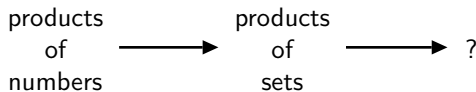
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Process of generalization for products:

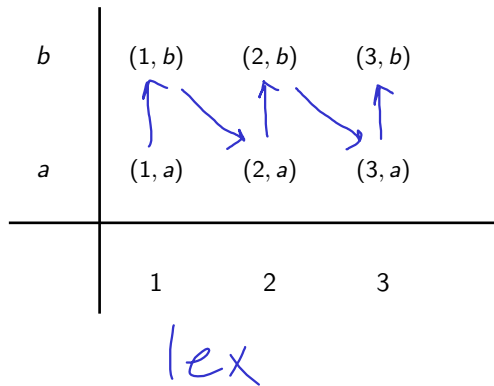


5. Products: ordered sets

Let's try making a definition of product for ordered sets.

11.

A is $1 \longrightarrow 2 \longrightarrow 3$, B is $a \longrightarrow b$

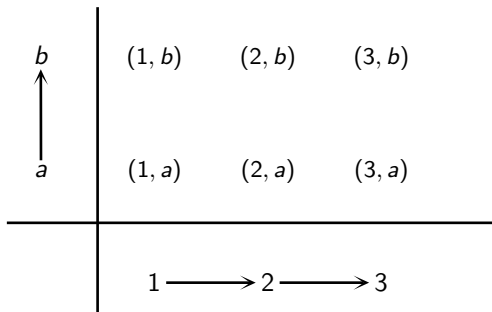


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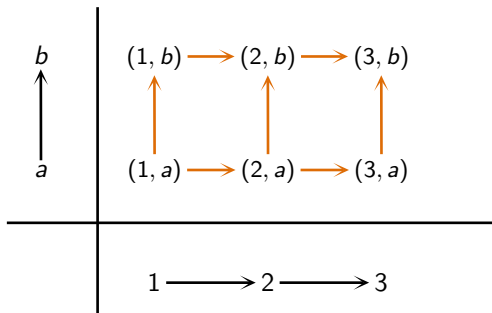


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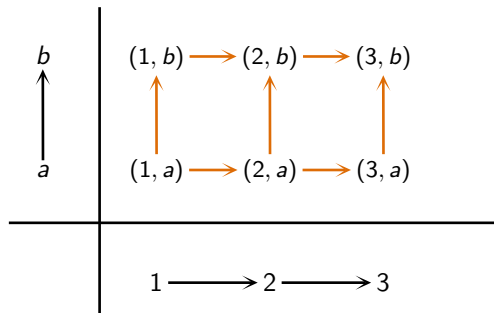


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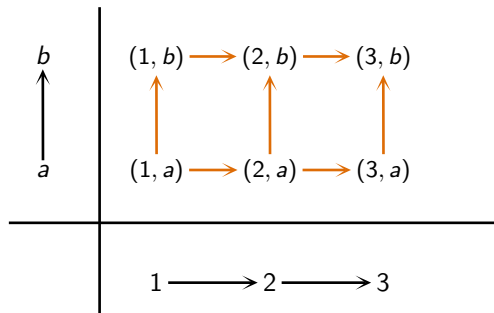


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Products of groups

We have seen an example of this: in $\mathbb{Z}_8$

$\times$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

5. Products: groups

Bed flipping

5. Products: groups

Bed flipping

	0	rot	flip	flop
0				
rot				
flip				
flop				

5. Products: groups

Bed flipping

	0	rot	flip	flop
0	0			
rot				
flip				
flop				

5. Products: groups

Bed flipping

	0	rot	flip	flop
0	0	rot		
rot				
flip				
flop				

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	0	rot	flip	flop
0	0	rot	flip	
rot				
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Bed flipping

	0	rot	flip	flop
0	0	rot	flip	flop
rot	rot			
flip	flip			
flop	flop			

Bed flipping

	0	rot	flip	flop
0	0	rot	flip	flop
rot	rot	0		
flip	flip			
flop	flop			

Bed flipping

	0	rot	flip	flop
0	0	rot	flip	flop
rot	rot	0	flop	
flip	flip			
flop	flop			

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	0	rot	flip	flop
0	0	rot	flip	flop
rot	rot	0	flop	
flip	flip			
flop	flop			

- Try filling in the rest of the table.
- How many Battenberg cakes do you see?

5. Products: groups

Bed flipping

	0	rot	flip	flop
0	0	rot	flip	flop
rot	rot	0	flop	flip
flip	flip	flop	0	rot
flop	flop	flip	rot	0



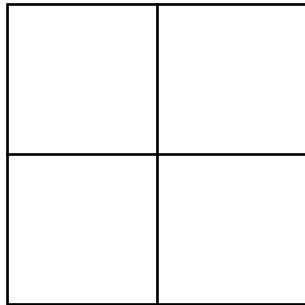
“Iterated Battenberg cake”

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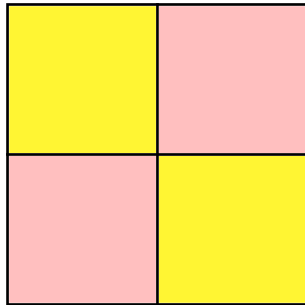
1	2
2	1

5. Products: groups

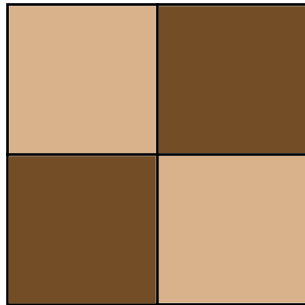


	2
2	

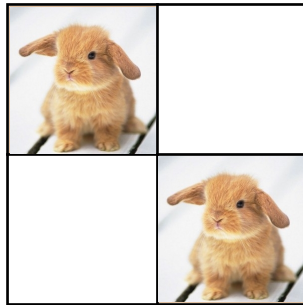
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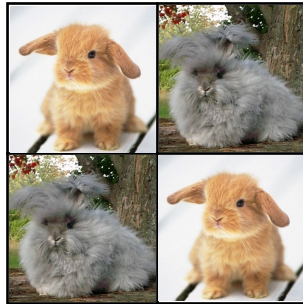
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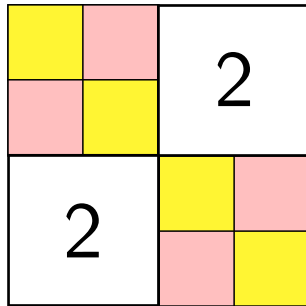


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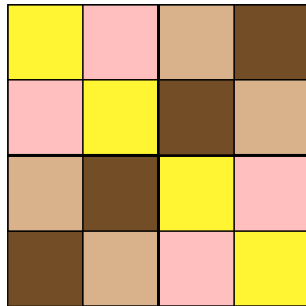


1	2
2	1

5. Products: groups



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5. Products: groups

$$\mathbb{Z}_2 \quad \begin{array}{c|cc} & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array}$$

$$\mathbb{Z}_2 \times \mathbb{Z}_2 = \{ (0,0), (0,1), (1,0), (1,1) \}$$



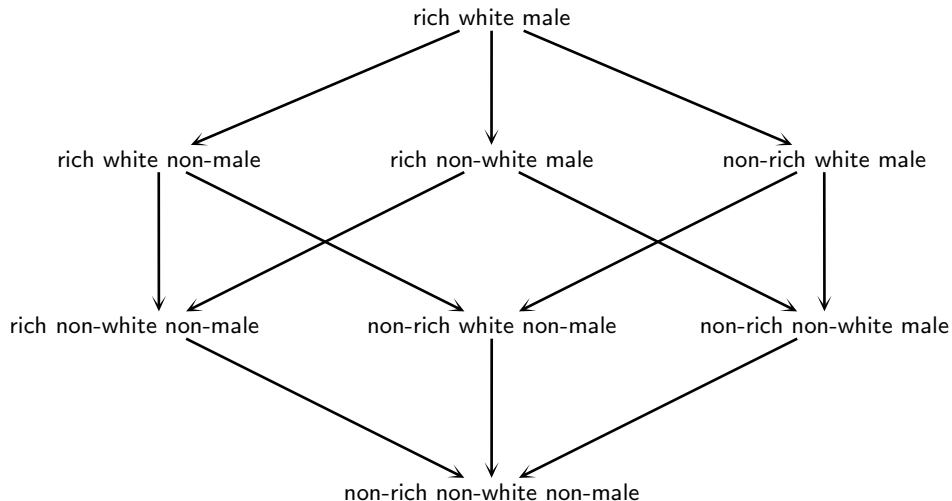
$(0,0)$	$(1,1)$	01	10	11
$(0,1)$	01	11	1,1	10
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6. Context

Something not universal in one place can be universal in another.

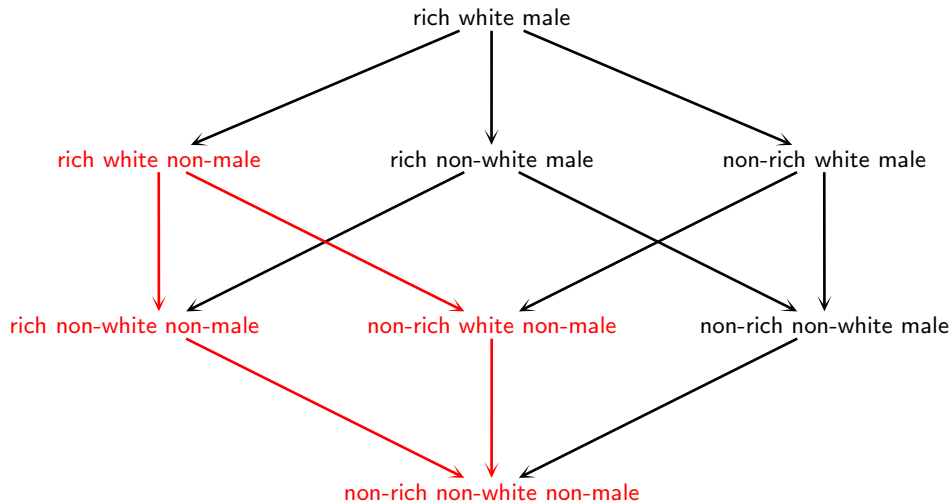
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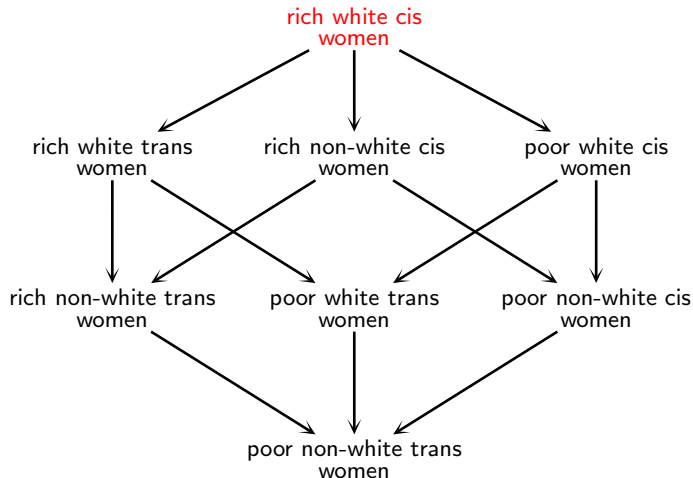


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6. Context



We can always restrict context to make something universal.

If you're the only person in the room then you're definitely the tallest. And the shortest.

Example: my life.

Conclusions

Further reading

- The ideas of category theory mostly covered in my first book *How to Bake π* .
- All the formality of what we covered, and more, is in my most recent book *The Joy of Abstraction*.
- Thoughts on social and political arguments are in *The Art of Logic in an Illogical World*.
- Thoughts on ingressive/congressive math and life are in *$x + y$: A Mathematician's Manifesto For Re-thinking Gender*.
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Thoughts to take away

- Abstraction has a point: it takes us further from “real life” but enables us to include a wider range of examples.
- So abstract math can be more relevant to our daily lives than applied math.
- Math is not just about solving problems; it is also about shedding light.
- I think the “shedding light” aspect of math is more congrressive and therefore can be more broadly inclusive.

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- All the formality of what we covered, and more, is in my most recent book *The Joy of Abstraction*.
- Thoughts on social and political arguments are in *The Art of Logic in an Illogical World*.
- Thoughts on ingressive/congressive math and life are in *$x + y$: A Mathematician's Manifesto For Re-thinking Gender*.
- My next book is *Is Math Real? How Simple Questions Lead us to Mathematics' Deepest Truths*.

Thoughts to take away

- Abstraction has a point: it takes us further from “real life” but enables us to include a wider range of examples.
- So abstract math can be more relevant to our daily lives than applied math.
- Math is not just about solving problems; it is also about shedding light.
- I think the “shedding light” aspect of math is more congrressive and therefore can be more broadly inclusive.

Category theory is a congrressive form of math that can be accessible and relevant to everyone.